

Kendy Application Admin Guide

Dockerized MERN Stack Management

Version 2.5 — May 5, 2025

Overview

This guide explains the architecture and administration of the Kendy MERN stack application running in Docker containers.

Quick Start

To launch the application:

```
docker compose up -d
```

To stop the application:

```
docker compose down
```

1. Database Administration

Accessing MongoDB

```
docker exec -it kendy-mongo-1 mongosh -u root -p root --authenticationDatabase admin
```

Essential MongoDB Commands

Command	Description
show dbs	List all databases
use KendyProject	Switch to application database
show collections	List all collections (tables)
db.users.find().pretty()	View formatted user data
db.stats()	Check database statistics
db.help()	Show available commands

2. Container Management

Daily Operations

Standard Workflow

1. **Start services:**

```
docker compose up -d
```

2. **Verify status:**

```
docker compose ps
```

3. **View logs:**

```
docker compose logs -f
```

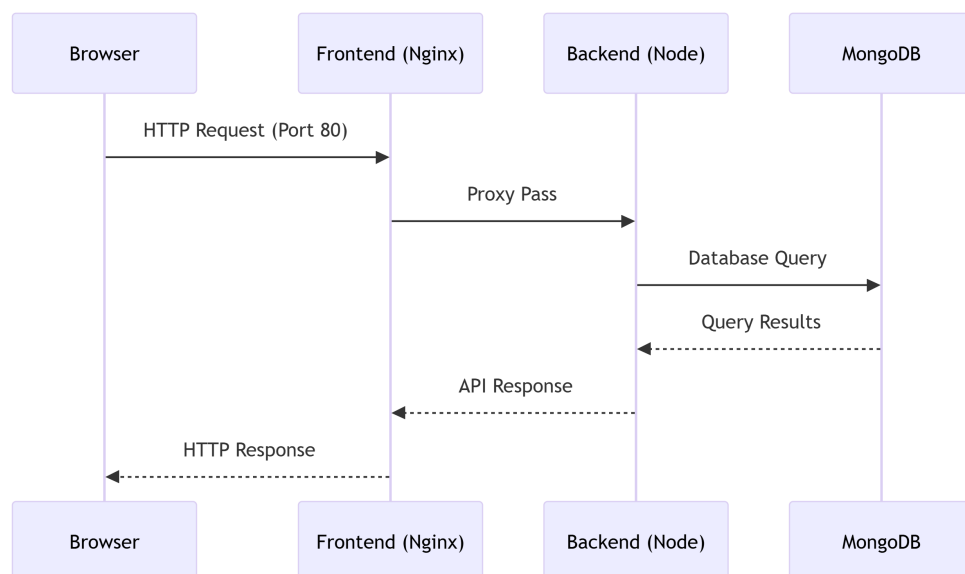
4. **Monitor resources:**

```
docker stats
```

Common Docker Commands

Command	Purpose
<code>docker compose restart</code>	Restart all services
<code>docker compose build</code>	Rebuild images
<code>docker exec -it backend bash</code>	Access backend container
<code>docker volume ls</code>	List data volumes

3. Application Architecture



Component Details

Frontend [Nginx + React](#)

- Serves static files on port 80
- Proxies API requests to backend

Backend [Node.js + Express](#)

- Runs on port 5001
- Connects to MongoDB
- Provides REST API and GraphQL

Database [MongoDB](#)

- Internal port 27017
- Data persisted in `mongo-data` volume
- Requires authentication

4. Troubleshooting Guide

Symptom	Solution
Containers won't start	Check <code>docker compose logs</code> for errors
502 Bad Gateway	Verify backend is running: <code>curl localhost:5001/health</code>
Database connection failed	Check MongoDB logs: <code>docker logs kendy-mongo-1</code>
Port conflicts	Identify used ports: <code>netstat -tuln</code> Or <code>lsof -i</code>

5. Maintenance Procedures

Regular Maintenance

Backup Database

```
docker exec kendy-mongo-1 mongodump \
-u root -p root \
--authenticationDatabase admin \
--db KendyProject \
--out /backup/$(date +%Y%m%d)
```

Update Procedure

1. Pull latest code changes
2. Rebuild containers:

```
docker compose up -d --build
```

3. Verify updates:

```
docker compose logs -f
```

Emergency Reset

Warning: This will delete all application data!

```
docker compose down -v
docker compose up -d
```

6. Used Resources

Resource	Details
Host Machine	Ubuntu Server 22.04 LTS (VM or bare-metal)
Virtualization Platform	VMware Workstation / VirtualBox / Proxmox
Containerization	Docker Engine 24+ with Docker Compose
Frontend Stack	React served by Nginx (Alpine variant)
Backend Stack	Node.js + Express with REST + GraphQL APIs
Database	MongoDB (official Docker image with persistent volume)
Reverse Proxy	Nginx (serving static files and forwarding API calls)